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TPE–401

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

PETROLEUM GEOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are reservoir traps ? What are the different types of traps ? Describe structural traps with neat sketches. (CO1, CO4, CO5)

(b) Explain source rock and why it is necessary for generation of H-C. Describe types of source rock and concept of potential, effective, relic effective Source Rock ? (CO1, CO4, CO5)

(c) Explain factor involved in the evaluation of hydrocarbon resources and reserves. (CO1, CO4, CO5)

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2. (a) Differentiate between paraffins, Cycloparaffins and aromatics and also give *three-three* examples. (CO1, CO2)

   (b) Describe in brief various types of porosity and permability found in carbonate and clastic sedimentary rock. (CO1, CO2)

   (c) Describe in brief about the petroliferous basins of India with 4 basins name in category I & II and 6 Basins name in category III. (CO1, CO2)

3. (a) What are 'biotic and abiotic' theories of origin of petroleum ? Discuss briefly. (CO1, CO3, CO4)

   (b) What do you mean by migration of hydrocarbon ? Explain primary, secondary migration and tertiary Migration. (CO1, CO3, CO4)

   (c) Describe type I, II, III and IV kerogen in petroleum. (CO1, CO3, CO4)

4. (a) What do you understand by TOC Explain, also write why TOC and Vitrinite Reflectance an Important Parameter. (CO2, CO4)

   (b) Why reservoir rocks are important in hydrocarbon exploration ? Describe any two types of reservoir rock. (CO2, CO4)

   (c) Describe the characteristics of source rock types and explain depositional environments in which they are typically deposited. (CO2, CO4)

5. (a) Briefly discuss the different types of structural and combination traps. (CO1, CO2)

   (b) Explain the types of information provided by Rock Eval pyrolysis testing of potential petroleum source rocks. (CO1, CO2)

   (c) Describe the favourable conditions for a rock formation to be a good petroleum reservoir. (CO1, CO2)

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END SEMESTER EXAMINATION JUNE- 2024

Name of The Program: ***B.Tech (Petroleum Engineering)*** Semester: **IV**

Name of The Course: **Reservoir Engineering-I** Course Code: **TPE-402**

Time: 3 Hours Maximum Marks: **100**

Note:

(i) All Questions are compulsory

(ii) Answer any two questions among a, b & c in each main question.

(iii) Total marks each main question is twenty.

(iv) Each Question carries 10 Marks

Q1. **(20 Marks)(CO1,3)**

a) **i)** Calculate the arithmetic average and thickness-weighted average from the following measurements.

SAMPLE	THICKNESS,ft	POROSITY,%
1	1.0	10
2	1.5	12
3	1.0	11
4	2.0	13
5	2.1	14
6	1.1	10

ii) Calculate average oil and connate water saturation from the following measurements:

Sample, %	hi, ft	Porosity, %	So, %	Swc, %
1	1.0	10	75	25
2	1.5	12	77	23
3	1.0	11	79	21
4	2.0	13	74	26
5	2.1	14	78	22
6	1.1	10	75	22

b) Explain the following

i. Laminar Flow
ii. Turbulent Flow
iii. Steady Flow
iv. Unsteady Flow
v. Uniform and Non-Uniform Flow
vi. Rotational Flow
vii. Irrotational Flow
viii. Viscous and Non-Viscous Flow

c) Describe Darcy law and also explain linear, Radial, spherical and hemispherical flow

Q2. **(20 Marks)(CO2,3)**

a) With the help of a table clearly explain the concept of grain size classification of clastic sediments.

b) Explain all physical properties of crude oil.

i. Total Porosity
ii. Effective Porosity
iii. Wettability
iv. Capillary Pressure.
v. GOR
vi. WOR
vii. Cloud Point
viii. Bubble Point
ix. Dew Point
x. Sweet Crude
xi. Sour Crude

Q3. **(20 Marks)(CO2,3)**

a) Classify crude oil on the basis of chemical composition.

b) Explain the concept that how fluid will flow through fractures, is it good for reservoir or not, give reason for your answer. Differentiate between Compressible and Incompressible flow.

c) With the help of a table explain Tissot classification

Q4. **(20 Marks)(CO3,4)**

a) Describe the concept of rock and fluid compressibility with types of compressibility.

b) Write a short note on
   i. Gas cap reservoir
   ii. Sorting
   iii. Explain cap rock and porosity and permeability and also differentiate between primary and secondary porosity

c) i) Calculate the pressure difference, i.e., capillary pressure, and capillary rise in an oil-water system from the following data:
Contact Angle = 30°
Water density = 1.0 gm/cm3
Oil density= 0.75 gm/cm3
r = 10-4 cm
Oil water surface tension = 25 dynes/cm

ii) A brine is used to measure the absolute permeability of a core plug. The rock sample is 4 cm long and 3 cm2 in cross section. The brine has a viscosity of 1.0 cp and is flowing a constant rate of 0.5 cm3/sec under a 2.0 atm pressure differential. Calculate the absolute permeability.

Q5 **(20 Marks)(CO3,4,5)**

a) With the help of a diagram explain the concept of water and gas conning.

b) Differentiate between Solution Gas Drive and gravity drive.

c) Explain the concept of water drive and gas cap drive with percentage of their recovery.

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END SEMESTER EXAMINATION 2024

Name of the Course: B.Tech Petroleum Engineering — *Semester: IV*

Name of the Paper: Heat and Mass Transfer — *Course Code:* TPE-403

Time: Three Hours — MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1		COs
a.	Deduce the equation for the logarithmic mean area for the hollow sphere. Solve: A Spherical shaped vessel of 1.4 m diameter is 90mm thick. Find the rate of heat leakage, if the temperature difference between the inner and outer surface is 220 ^{0}C. Thermal conductivity of the material of the sphere is 0.083 W/m^{0}C.	CO1, CO2
b.	Define insulation and hence critical radius of insulation? What are the applications of insulation? Derive an expression for Critical radius of insulation for cylindrical pipe. What is the ratio of critical radius of insulation of sphere and cylinder?	CO1, CO2
c.	A steam pipe of outer diameter 120mm is covered with two layer of lagging, inside layer 45 mm thick (k = 0.08 W/m^{0}C) and outer layer 30mm thick (k = 0.12 W/m^{0}C). The pipe conveys steam at a pressure of 20 bar with 50^{0}C superheat. The outside temperature of lagging is 25^{0}C. If the steam pipe is 30m long, determine: i. Heat lost per hour ii. Interface temperature of lagging The thermal resistance of the steam pipe may be neglected.	CO3
Q2		
a.	Discuss free convection from a flat surface for a vertical plate, hot or cold suggested by Churchill and Chu. Is this equation applicable for an inclined plane if **yes** then what should be the minimum angle of inclination? If **No** why?	CO3, CO4
b.	Air flows over a flat surface, 2m in length oriented in the direction of flow and of sufficient breadth, maintained at 150^{0}C. The pressure is 1 atm and the bulk air temperature is 30^{0}C. If the air velocity is 12 m/s, determine (i) the local heat transfer coefficient as a function of longitudinal position, (ii) the average heat	CO1, CO4

	transfer coefficient and (iii) the rate of heat loss from the surface. The relevant physical properties of air are taken as: **$\rho = 0.962$ kg/m^3, $\mu = 2.131 \times 10^{-5}$Kg/m.s, K = 0.031 W/m^{0}C, $C_p = 1.01$ Kj/kg ^{0}C**	
c.	Define the following with their physical significance and their mathematical expressions 1. Reynolds no. 2. Nusselt no. 3. Prandatl no. 4. Radisity (J). 5. Irradiation.(G)	CO1, CO2
Q3		
a.	Define absorptivity, reflectivity and transmissivity for the incident radiation and hence explain the following: i. Black body ii. Opaque body iii. White body iv. Gray body	CO1
b.	Define View/Shape factor? Define interchange factor for the radiant heat exchange between infinite long concentric cylinders. Determine the radiant heat exchanger in W/m^2 between two large parallel steel plates of emissivity's 0.8 and 0.5 held at a temperature of 1000 K and 500 K respectively, if a thin copper plate of emissivity 0.1 introduced as a radiation shield between the two plates. Use $\sigma = 5.67 \times 10^{-8}$ W/m^2 K^4	CO1, CO2
c.	Derive an expression for the heat exchange between the nonblack bodies for infinite parallel planes mentioning the assumptions made for the analysis of heat exchange between two nonblack parallel surfaces. Specify the symbol used.	CO2, CO5
Q4		
a.	Discuss and derive the electrical network analogy for thermal radiation systems and hence explain the Surface resistance and space resistance?	CO1, CO5
b.	Write the assumptions made during the design aspects of heat exchanger. Derive the relation of logarithmic mean temperature difference [LMTD] for counter flow heat exchanger. Specify the symbol used.	CO4, CO5
c.	Discuss the different types of heat exchanger and their classifications on the various aspects? Solve: A shell and tube heat exchanger is used for cooling crude oil from 400 K to 360 K. Crude oil flows through the tube at 3650 kg/h. Water enters the shell side at 310 K and has a flow rate of 1600 kg/h. Assume the heat capacity of crude oil and water as 2.5 kJ/kg.K and 4.187 kJ/kg.K, respectively. If the overall heat transfer coefficient is 300 Wm2.K/ and the streams are countercurrent, the heat transfer area required is _____ m2 (rounded to one decimal place).	CO3, CO6

Q5		
a.	Illustrate the modes of Mass Transfer. Define the mass transfer terms with the mathematical expression of i. Concentrations ii. Velocities iii. Fluxes	CO4, CO6
b.	Define the laws for the mass transfer? What is mass transfer coefficient? Derive an expression for the steady state diffusion through a plain membrane	CO1, CO5
c.	Define mass transfer by convection. Define diffusional resistance and hence Derive the expression for steady state diffusion through a cylindrical shell.	CO3, CO4

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Program: **B. Tech. Petroleum Engineering** Semester: **IV**

Name of the Course: **Applied Sedimentology** Course Code: **TPE – 404**

Time: **3 Hours** Maximum Marks: **100**

Note:

i) All questions are compulsory.
ii) Answer any two sub questions among **a, b & c** in each main question.
iii) Total marks for each main question are twenty.
iv) Each questions carries **10 Marks**

Q1		**(20 marks)**
(a)	Differentiate between Erosion and Weathering? Explain principal diagenesis processes and changes that occur in siliciclastic sedimentary rocks during burial.	CO1,
(b)	Illustrate how sedimentary grain sorting of a rock influences reservoir qualities of rocks.	CO1, CO3
(c)	Illustrate the relationship between fluid properties (density and viscosity) and the Reynolds number, and explain its relevance in predicting flow regimes.	CO3
Q2		**(20 marks)**
(a)	Describe the role of sediment transport in the formation of clastic sedimentary rocks.	CO4
(b)	Describe how the distance over which sediment is transported influences the size, roundness, and sphericity of sediment particles.	CO2
(c)	Examine the importance of porosity and permeability in the context of petroleum exploration.	CO5
Q3		**(20 marks)**
(a)	Explore the concept of bedding planes in sedimentary rocks. Explain how bedding planes form and why they are important in understanding the layering of sedimentary rocks.	CO5
(b)	Illustrate principles of stratigraphy and their significance.	CO2
(c)	Describe the various sedimentary structures with help of proper diagrams.	CO2, CO4
Q4		**(20 marks)**
(a)	Discuss the sandstone maturity and different types of sandstone classification.	CO3
(b)	Discuss about various types of sedimentary depositional environments?	CO4
(c)	Describe the physical and chemical parameters of depositional environments. Discuss their role in sedimentary rock formation.	CO5
Q5		**(20 marks)**
(a)	Discuss the Fore-arc basin and Pull-apart basin including their tectonic setting, sedimentary processes and types of sediments.	CO5
(b)	Discuss the classification of sedimentary basins based on their tectonic setting, including rift basins, foreland basins, and intracratonic basins.	CO3, CO4
(c)	Elaborate the classification of non-clastic sedimentary rocks with suitable examples	CO6

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END SEMESTER EXAMINATION JUNE -2024

Name of the Course: B. Tech Petroleum Engineering — Semester: 4th

Name of the Paper: Petroleum Drilling Engineering I — Course Code: TPE 405

Time: 3 Hours — Maximum Marks: 100

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b and c in each main question.

(iii) Total marks in each main question is twenty.

Q1		**(20 marks)**	
(a).	What are the objectives of well planning? Describe the parameters to be considered in well planning for a vertical well?		CO1
(b)	The following data refer to an oilwell block-and-tackle system: Number of lines = 10 (with EF = 0.81) Maximum expected hook load = 500,000 lbf Hook load speed = 120 ft/min Hoisting drum diameter = 32" Mechanical efficiency of draw works = 0.88 Calculate 1. The power at the drawwork 2. The motor power required 3. The fastline 4. Motor to drum gear ratio when pulling out of hole the maximum allowable load. Note:Use an efficiency factor of 0.81.		CO2
(c)	What is the purpose of hosting system in drilling operations? Describe the main components of hosting system used in drilling.		CO1
Q2		**(20 marks)**	
(a)	What are drilling fluids, and what are their primary functions in drilling operations?		CO3
(b)	Describe common drilling problems associated with drilling fluid, such as wellbore instability, stuck pipe, lost circulation, and differential sticking.		CO3
(c)	Explain the concept of viscosity in drilling fluids and its role in fluid flow behavior, pumpability, and hole cleaning efficiency.		CO3
Q3		**(20 marks)**	**CO**
(a)	What are the main types of casing used in drilling operations, and how do they differ in terms of size, design, and application?		CO4
(b)	What is the purpose of cementing in oil and gas wells, and why is proper cement placement essential for well integrity and production?		CO4

(c)	Explain the function of retarders in cement slurries and how they help to extend the thickening time and improve cement placement in deep or high-temperature wells.	CO4
Q4	**(20 marks)**	
(a)	Explain how hydrostatic pressure helps to counterbalance formation pressures encountered during drilling and prevent blowouts or fluid influxes into the wellbore.	CO5
(b)	What is laminar flow, and how does it differ from turbulent flow in drilling operations? Discuss the significan of laminar and turbulent flow during drilling.	CO5
(c)	What is surge pressure in the context of drilling operations, and how does it affect wellbore stability and drilling efficiency?	CO5
Q5	**(20 marks)**	
(a)	Discuss the differences between fixed cutter (diamond) drill bits and roller cone (tricone) drill bits in terms of design, materials, and drilling performance.	CO6
(b)	Discuss the factors affecting bit bear during drilling.	CO6
(c)	Describe the main factors that influence penetration rate during drilling.	CO6

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END SEMESTER EXAMINATION 2024

Name of the Course: B.Tech Petroleum Engineering *Semester: IV*

Name of the Paper: Mud Logging *Course Code:* TPE-413

Time: Three Hours MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1		COs
a.	What is the difference between mud logging and Well logging? What are its limitations?	CO-1
b.	What is the mud logging principle? Describe different component of Mud logging Units. What are the various sensors available in recording the data?	CO-1
c.	What is Rig up and rig down in mud logging? What is the precaution to be taken during this process?	CO-2
Q2		
a.	What is Lag up, lag-Down, and complete cycle ? Explain in details. How is the mud logging is useful in Formation Evaluation?	CO-2
b.	What is Lag time and Lag strokes? Why it is important? How is it calculated?	CO-2
c.	What are the various sensors used for recording mud logging data? What are the various mud logging parameters recorded during data acquisition?	CO-3
Q3		
a.	Explain drilling fluids & Its importance. How many type of drilling fluid are uses during drilling operation?	CO-5
b.	What are the uses of drilling cutting analysis during mud logging? What are the reservoir parameters estimated from cutting data?	CO-3
c.	Explain the Coring and coring methods. Why it is requires? What are the reservoir parameters measured from coring?	CO-4
Q4		
a.	How many type of coring are taken during drilling process? Explain each in details. How the core leveling is done at well site? Draw figure.	CO-4
b.	What are wet cut & dry cut sample? What are the information recorded in these sample?	CO-4
c.	Write the core handling process in details. What are the precaution required for core handing?	CO-4

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Q5		
a.	Explain the reservoir. What are the main parameters required for a good reservoir? What is the difference between conventional gas reservoirs and CBM gas reservoir?	CO-5
b.	Explain gas hydrates. How they are formed? What are the essential condition for gas hydrate?	CO-5
c.	Explain Hydrostatic pressure, Pore pressure, abnormal pressure. Over pressure, Under pressure in details. Why they are important in oil industry?	CO-5

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End Semester Examination 2024

Name of the Program: B.Tech. (Petroleum Engineering)

Semester: 4th

Name of the Course: Numerical and Geostastical Methods

Course Code: TMA 403

Time: 3 Hours **Maximum Marks:** 100

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub question carries 10 marks

Q1	(20 marks)	
a)	Using Bisection method to compute a real root of the equation $x^2 = 13$, (Correct to three decimal places).	
b)	Solve the following equations by Gauss's elimination method: $2x - y + 3z = 9$ $x + y + z = 6$ $x - y + z = 2$	CO-1, 2
c)	Using Newton's method, find a real root of an equation $xe^x = 1$ correct to three decimal places between 0 and 1.	
Q2	(20 marks)	
a)	Using Trapezoidal rule, find the value of the integral $\int_4^{5.2} \log x \, dx$ (Accuracy correct to five decimal places) for the following table:	CO-3, 4

x	4	4.2	4.4	4.6	4.8	5.0	5.2
$y = \log x$	1.38629	1.43508	1.48160	1.52605	1.56861	1.60943	1.64865

b) Evaluate the integral $\int_0^6 \sqrt{x}\, e^{-x} dx$ using Simpson's one-third rule by taking 6 equal subintervals.

c) Using Euler's method to find $y(0.4)$ correct to 3 decimal places. Given that the initial value problem $\frac{dy}{dx} = x - y^2$ with initial condition $y(0.2) = 0.02$ and step size $h = 0.2$.

Q3 (20 marks)

a) The following data representant Carbon dioxide emission from coal-fluid boilers (in units' of 1000 tons) over a period of years 1965, and 1977. The independent variable (years) has been standardized yield the following table:

Years(x)	0	5	8	9	10	11	12
CO2 emission (y)	910	680	520	450	370	380	340

To fit the simple linear regression line using the above data.

b) Solve the following system of equations: $x + y - z = 0,\ -x + 3y = 2,\ x - 2z = 3,$ by the Gauss–Seidel method. Assume the initial solution $X = (0.8, 0.8, 2.1)^t$. CO-1, 5

c) The following are the values of two variables, diastolic blood pressure (y) and time in minutes (x). Diastolic blood pressure (y) was recorded at various time periods (x) on individuals. Find the regression line(s) of y on x and x on y also find the coefficient of correlation:

Time periods (x)	0	5	10	15	20
Diastolic blood pressure (y)	72	67	70	65	66

Q4 (20 marks)

a) Using Least square method to fit a parabolic curve $y = ax^2 + bx + c$ for the following data:

x	1	3	5	10	15
y	10	5	20	25	22

CO-6

b) Define Correlation and Regression analysis with the help of one example.

c) Following data depicts the statistical values of rainfall and production of wheat in a region for a specified time period.

	Mean	Standard Deviation
Production of Wheat (Kg/unit area	10	8
Rainfall (cm.)	8	2

Estimate the production of wheat when rainfall is 9 cm if correlation coefficient between production and rainfall is given to be 0.5.

Q5 (20 marks)

a) Explain the terms spatial analysis, data posting, contour map, Indicator map

b) What is geostatistics? How it is useful for geological work?

c) Explain the term Kriging. How Kriging is used in estimation of values in unknown places using deterministic & probabilistic simulation technique.

CO-4, 5